An Analysis of the Serine Protease Family

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INTRODUCTION

Serine proteases are a large family of enzymes that have a wide range of functions in the body including blood coagulation, digestion, immunity, inflammation and development [2]. 99% of all serine proteases have a catalytic triad made up of His-57, Asp-102 and Ser-195, and an oxyanion hole which are the residues mainly responsible for their enzymatic action. In addition, serine-214 (alignment index 398) which was 98% conserved in this alignment, is sometimes considered the fourth member of the catalytic triad. There are four subgroups of serine proteases, chymotrypsin, subtilisin, carboxypeptidase Y, and Clp proteases.

Mechanistically, most serine proteases either activate the amide bond or distort the peptide bond and they always protonate the amine bond before expulsion. They also always activate water and increase specificity by increasing k_{cat}, which is controlled mainly by the S1 site. Another part of the mechanism of serine proteases is the oxyanion hole (formed in part by Gly-193 and Ser-195), which activates the carbonyl of the scissile peptide bond by stabilizing the oxyanion by forming a positively-charged pocket [1].

The S1 specificity site is made up of the residues 189-192, 214-216, and 224-228 with substrate specificity mainly determined by residues 189, 216, and 226 . The S1 site of chymotrypsin-like serine proteases mainly determines substrate specificity as the S2 and S3 sites have wide ranges of substrates that can bind to them. The S2 and S3 sites help determine substrate specificity more in other serine proteases such as elastase and enteropeptidase [1].

In this study, the goal was to understand better how the specific amino acid sequences of each protein relates to function and contribute to knowledge of serine proteases as a whole. To do this, an alignment of 4200 different serine protease enzymes was made and analyzed. Three protein groups within the serine protease family (trypsin, granzyme B, and clotting factor IX) were then compared in more depth in order to find similarities and differences in their sequences that contribute to their unique functions. Granzyme B is a chymotrypsin-like serine protease that helps control apoptosis by cytotoxic lymphocytes [3], clotting factor IX is a zymogen enzyme involved in the blood coagulation cascade [2], and trypsin is a digestive protease found in the pancreas [1]. Travel funding from ISCB was obtained for this presentation.

FIGURE 1

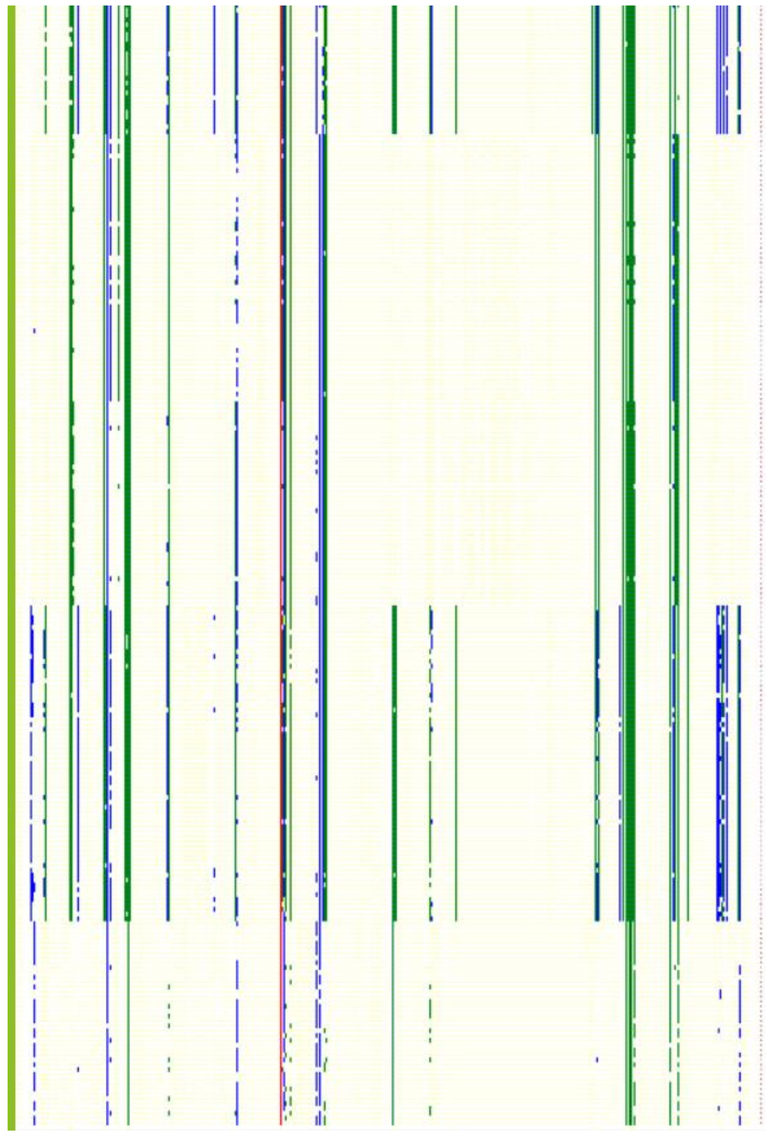

Figure 1: Alignment of 4200 different serine protease amino acid sequences generated by T-Coffee and manually adjusted using tertiary structures as guides. Amino acid conservation is shown as follows: red columns of the alignment indicate amino acids at 100% conservation (1 residue), green indicates 80-99% conservation (44 residues), blue areas represent 60-79% conservation (25 residues), and white are not conserved overall.

FIGURE 2

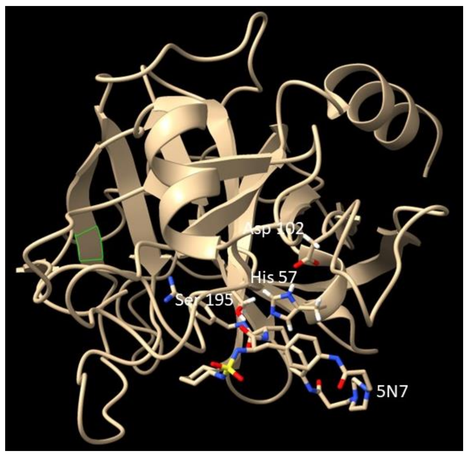

Figure 2: View of the catalytic triad in bovine trypsin (PDB ID 5EG4) with a cyclic inhibitor bound. The alignment has revealed conservation of the catalytic triad residues Histidine-57 (index 61), Serine-195 (370) and Aspartate-102 (155) in at least 99% of all the sequences.

FIGURE 3

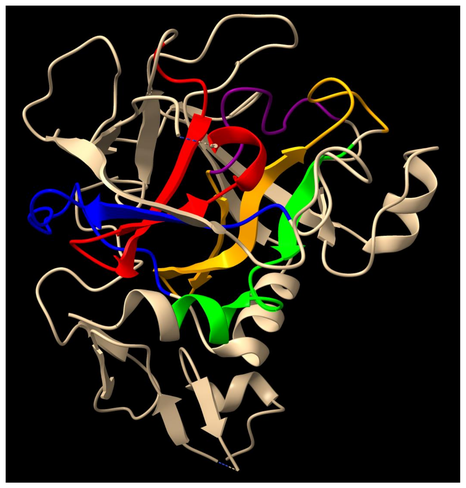

Figure 3: View of the position of the most well conserved sequence motifs identified by MEME [5] in human clotting factor IX (PDB ID 6X5L). Motif 1, which contains His-57 (61), is colored red. Motif 2 convers part of the S1 substrate site. Motif 3 is green. Motif 4 is purple and contains part of the S1 substrate site, Ser-195 (370), and the oxyanion hole. Motif 5 is blue and contains Asp-102 (155). In bovine trypsin (not shown) the ligand is coordinated by motifs 1 and 4. The green and the blue motifs (3 and 5 respectively) are more structural.

FIGURE 4

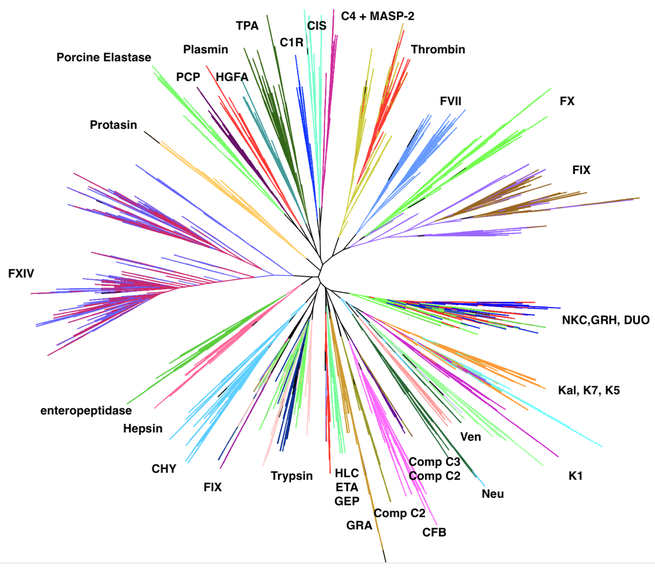

Figure 4: Bootstrapped phylogenetic tree produced with neighbor-joining algorithm [6] for the serine protease family. Being an unrooted tree, it does not indicate evolutionary origin, but rather sequence relatedness. Several clotting factors are clustered together, although not all. A similar phenomena occur for some of the complement proteins as well as digestive proteases.

FIGURE 5

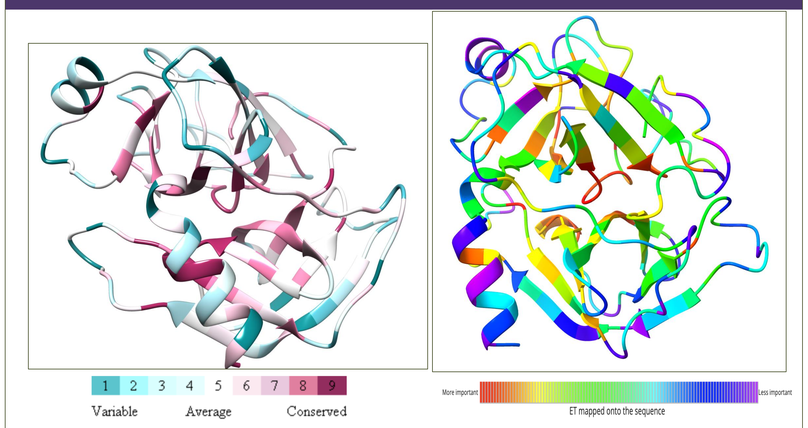

Figure 5: Consurf analysis [4] (left) and Evolutionary Trace Analysis [7] (right) of amino acid conservation in bovine trypsin. Most conservations center around the catalytic triad and oxyanion hole in the enzyme's active site.

FIGURE 6

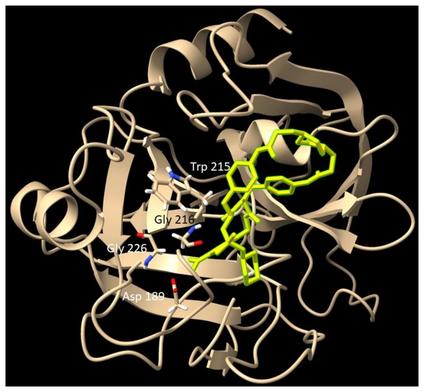

Figure 6: The residues pictured here in bovine trypsin were found to be group-specific conservations (as identified by Evolutionary Trace [7]) in trypsin, granzyme B, and clotting factor IX. These four residues all contribute to each enzyme's unique substrate specificity. The inhibitor 5N7 is colored yellow in this image.

CONCLUSIONS

- Most conserved amino acids in this family are in areas including and surrounding the catalytic triad, oxyanion hole, and substrate specificity sites. There were also conservations found that were critical for structural folding.
- Glycines were overly represented among the conserved residues (11 of 44 of at least 80% conservation). This is due to their lack of side chain and ability to participate in protein folding. This is also true for several other protein families [8].
- In the phylogenetic tree many similar types of proteins were grouped together, such as the kallikreins, complement proteins and clotting factor proteins. However, this was not true for all of the proteins in these groups.
- Glutamyl enteropeptidase (GEP) and exfoliative toxin A (ETA) had the least amount of amino acid conservations, e.g. were most divergent of the serine proteases studied.
- The residue at index 424 influences whether there is a sodium bound to that enzyme. If this position is a proline, there is no sodium bound. This is the case in enzymes like trypsin and granzyme B. If this position is a tyrosine, there is a bound sodium, which is the case in clotting factor IX.
- The residue at index 366 of the alignment is 93% conserved cysteine and usually forms a disulfide bond with the cysteine at index 406, but several proteases including clotting factor IX, NK cell protease, and duodenase lack this disulfide.
- It was found that residues at indices 374 and 397 are both important for hydrophobic packing. However, His-204 (374) in clotting factor IX forms a group-specific hydrogen bond to Thr-215 (394) are important for maintaining structure.
- Some variable positions within the alignment are involved with group-specific substrate specificity, especially indices 364, 399, 400, 425, and 427, which are all in the S1 binding site in granzyme B, factor IX and trypsin.
- In a compensatory mutation if the residue at index 220 is a cysteine or aliphatic (V/I/L) residue, the corresponding residue at index 374 is also an aliphatic residue. If the residue at index 220 is a glycine, the the residue at index 374 is either an aliphatic or aromatic (F/Y) residue. If the residue at index 220 is a serine, then it forms a hydrogen bond with a histidine at index 374.

LITERATURE CITED

1. Hedstrom, Lizbeth. "Serine protease mechanism and specificity." *Chem. Rev.* 102 (2002): 4501-4523.
2. Sorensen, Anders, et al. "Beating tissue factor at its own game: design and properties of a soluble tissue factor- independent coagulation factor VIIa." *J. Biol. Chem.* 295(2) (2020): 517-528.
3. Waugh, Sandra. "The structure of the pro-apoptotic protease granzyme B reveals the molecular determinants of its specificity." *Nature Structural Biology* 7.9 (2000): 762-765.
4. Glaser F., Pupko T., Paz I., Bell R.E., Bechor D., Martz E. and Ben-Tal N. 2003. ConSurf: Identification of Functional Regions in Proteins by Surface-Mapping of Phylogenetic Information. *Bioinformatics* 19:163-164.
5. Bailey TL, Elkan C. Fitting a mixture model by expectation maximization to discover motifs in biopolymers. ISMB. 1994; 2: 28–36. PMID: 7584402
6. Felsenstein J. PHYLIP manual, Version 3.3. Berkeley: University of California, University Herbarium. 1990.
7. Lichtarge O, Bourne HR and Cohen FE. An evolutionary trace method defines binding surfaces common to protein families. *J Mol Biol.* 1996 Mar; 257(2):342-58.
8. Clark L, Leatherby D, Krilich E, Ropelewski AJ, Perozich J. In silico analysis of class I adenylate-forming enzymes reveals family and group-specific conservations. PLoS One. 2018 Sep 4;13(9):e0203218. doi: 10.1371/journal.pone.0203218. PMID: 30180199; PMCID: PMC6122825.

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